

Multiple Choice Questions – Lecture 1 (Software Engineering)

1. Which of the following best defines software?

- A) A single executable program
- B) Hardware plus operating system
- C) A collection of programs, configuration files, and documentation
- D) Just source code

Answer: C

2. If a program is written for personal use only, which of the following is least necessary?

- A) User documentation
- B) Program design documentation
- C) Usage instructions
- D) All of the above

Answer: D

3. Which of the following is an example of generic software?

- A) Air traffic control system
- B) Payroll system for a company
- C) Microsoft Word
- D) A custom-built hospital database

Answer: C

4. In customized software, who controls the specification?

- A) Developers
- B) Customers
- C) Both developers and customers
- D) End users only

Answer: B

5. Which attribute of good software ensures that the system is understandable and compatible with other systems?

- A) Dependability
- B) Acceptability
- C) Maintainability
- D) Efficiency

Answer: B

6. Which of the following is NOT part of software dependability?

- A) Reliability
- B) Security
- C) Safety
- D) Compatibility

Answer: D

7. Which attribute ensures that a software product can evolve to meet changing needs?

- A) Efficiency
- B) Maintainability
- C) Security
- D) Usability

Answer: B

8. Software engineering is defined as:

- A) The science of programming only
- B) The discipline concerned with the development and maintenance of software products
- C) Writing code without documentation
- D) Only managing software projects

Answer: B

9. Which of the following was the cause of the Ariane 5 disaster?

- A) Hardware malfunction
- B) Integer overflow due to reused software
- C) Fuel miscalculation
- D) Incorrect rocket design

Answer: B

10. Which of the following is NOT a reason why software engineering is important?

- A) National infrastructures depend on software
- B) Almost everyone uses mobile phones with software
- C) Entertainment is software-intensive
- D) Hardware has become more expensive than software

Answer: D

11. Which factor drives the need for software engineering because user requirements evolve constantly?

- A) Scalability
- B) Rapid change
- C) Cost
- D) Quality management

Answer: B

12. Which factor explains why large, complex systems need systematic approaches?

- A) Cost
- B) Scalability
- C) Large software size
- D) Dynamic nature

Answer: C

13. Which factor highlights the difficulty of scaling up software without systematic processes?

- A) Scalability
- B) Cost
- C) Dynamic nature
- D) Quality management

Answer: A

14. Which process activity involves defining functionality and constraints of software?

- A) Development
- B) Validation
- C) Specification
- D) Evolution

Answer: C

15. Which process activity ensures the software does what the customer wants?

- A) Specification
- B) Development
- C) Validation
- D) Evolution

Answer: C

16. Which is the main difference between computer science and software engineering?

- A) Software engineering is more theoretical
- B) Computer science is more practical
- C) CS deals with theories, SE with practical development problems
- D) There is no difference

Answer: C

17. System engineering differs from software engineering in that:

- A) It includes hardware and process design as well
- B) It only deals with coding
- C) It ignores hardware
- D) It focuses only on embedded systems

Answer: A

18. Which of the following is NOT a type of software application?

- A) Stand-alone applications
- B) Embedded control systems
- C) Batch processing systems
- D) Firmware-only applications

Answer: D

19. Which application type is most concerned with high-quality interaction and user experience?

- A) Batch processing
- B) Entertainment systems
- C) Data collection systems
- D) Stand-alone applications

Answer: B

20. Software in washing machines belongs to which application type?

- A) Interactive transaction-based applications
- B) Embedded control systems
- C) Stand-alone applications
- D) Batch processing systems

Answer: B

21. E-commerce platforms belong to which application type?

- A) Batch processing systems
- B) Interactive transaction-based applications
- C) Systems of systems
- D) Modeling and simulation

Answer: B

22. Payroll systems are examples of:

- A) Batch processing systems
- B) Modeling and simulation
- C) Embedded systems
- D) Interactive applications

Answer: A

23. Which application type often requires high-performance parallel computing?

- A) Stand-alone
- B) Data collection
- C) Modeling and simulation
- D) Batch processing

Answer: C

24. Collecting sensor data from engines represents which type of application?

- A) Systems of systems
- B) Embedded systems
- C) Data collection and analysis systems
- D) Stand-alone applications

Answer: C

25. Enterprise Resource Planning (ERP) is an example of:

- A) Batch system
- B) Stand-alone application
- C) System of systems
- D) Interactive transaction-based application

Answer: C

26. Which fundamental principle applies to all software systems?

- A) They should only be built from scratch
- B) They must follow a managed process
- C) They should not require documentation
- D) They must be entertainment-based

Answer: B

27. Which is a major challenge in software engineering due to operating across diverse devices?

- A) Scalability
- B) Security
- C) Heterogeneity
- D) Evolution

Answer: C

28. Legacy systems increase which challenge?

- A) Business and social change
- B) Heterogeneity
- C) Cost
- D) Quality management

Answer: B

29. Rapidly changing technologies emphasize which challenge?

- A) Business and social change
- B) Scalability
- C) Maintainability
- D) Efficiency

Answer: A

30. Which challenge emphasizes the importance of preventing malicious attacks?

- A) Scalability
- B) Business change
- C) Security and trust
- D) Heterogeneity

Answer: C

31. Respecting confidentiality of clients is part of:

- A) Technical competence
- B) Software validation
- C) Software engineering ethics
- D) Documentation

Answer: C

32. Which ethical principle emphasizes not misrepresenting skills?

- A) Confidentiality
- B) Competence
- C) Computer misuse
- D) Intellectual property rights

Answer: B

33. Which ethical issue concerns malware distribution?

- A) Confidentiality
- B) Competence
- C) Intellectual property
- D) Computer misuse

Answer: D

34. Which case study represents an embedded system?

- A) Medical records system
- B) Insulin pump controller
- C) Weather station
- D) Learning environment

Answer: B

35. Which case study focuses on data integrity and security?

- A) Insulin pump
- B) Medical records system
- C) Weather station
- D) Learning environment

Answer: B

36. Which case study represents a sensor-based system?

- A) Medical records system
- B) Digital learning environment

- C) Weather station
- D) Insulin pump controller

Answer: C

37. Which case study represents a support environment?

- A) Weather station
- B) Learning environment
- C) Medical records system
- D) Insulin pump controller

Answer: B

38. Which factor does software engineering help reduce by avoiding rework?

- A) Cost
- B) Quality
- C) Security
- D) Dynamic nature

Answer: A

39. Which factor ensures that the final product is reliable and user-friendly?

- A) Quality management
- B) Scalability
- C) Rapid change
- D) Dynamic nature

Answer: A

40. Which activity involves modifying software to meet changing customer needs?

- A) Validation
- B) Specification

- C) Development
- D) Evolution

Answer: D

Wish you all the best!

Fathy.